

Aniruddha Rakshit

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EDUCATION

State University of New York (SUNY) Binghamton

Ph.D. Candidate in Computer Science, Advisor: Jayson Boubin

Binghamton, NY

August, 2022–August, 2027 (expected)

Jahangirnagar University

M.S. in Computer Science and Engineering

Dhaka, Bangladesh

April 2015–February 2017

Jahangirnagar University

B.S. in Computer Science and Engineering

Dhaka, Bangladesh

2010–2014

SKILLS

- **Programming Languages:** Python, C++, C
- **Machine Learning Libraries:** PyTorch, Tensorflow, Keras, Scikit-Learn, Pandas, Matplotlib
- **Simulation and Design Tools:** PyBullet, MATLAB, ROS
- **System Design Skill:** Docker, Kubernetes, Minikube, Microsoft Azure
- **Embedded Device:** Raspberry Pi, NVIDIA Jetson Orin Nano and Xavier, ESP32

PUBLICATIONS

- [1] S. Das, J. Cheng, **Aniruddha Rakshit**, B. Soni, J. Boubin, and R. Ramnath, “Spice: Structured pruning for inference on constrained edge devices”, in *2026 IEEE 23rd Consumer Communications Networking Conference (CCNC)*, 2026, pp. 1–6.
- [2] S. Das, J. Cheng, **Aniruddha Rakshit**, J. Boubin, and R. Ramnath, “Epic: Efficient pruning for inference on constrained devices”, in *Practice and Experience in Advanced Research Computing 2025: The Power of Collaboration*, ser. PEARC '25, Association for Computing Machinery, 2025, ISBN: 9798400713989.
- [3] **Aniruddha Rakshit**, S. Reddy, R. Ramnath, A. Arora, and J. Boubin, “Righteous: Automatic right-sizing for complex edge deployments”, in *2024 IEEE/ACM Symposium on Edge Computing (SEC)*, 2024, pp. 15–28.
- [4] R. Hafiz, M. R. Haque, **Aniruddha Rakshit**, and M. S. Uddin, “Image-based soft drink type classification and dietary assessment system using deep convolutional neural network with transfer learning”, in *Journal of King Saud University - Computer and Information Sciences*, 2022, pp. 1775–1784.
- [5] F. Ahmed, A. R. Choudhury, **Aniruddha Rakshit**, and M. Z. Hasan, “2020 iee region 10 symposium (tensymp)”, in *An IoT Based System for Printing Braille Letter from Speech*, 2020, pp. 344–347.

WORK EXPERIENCE

State University of New York at Binghamton

Graduate Research Assistant

Binghamton, USA

Spring 2026- Present

State University of New York at Binghamton

Graduate Assistant/ Teaching Assistant

Binghamton, USA

Fall 2022-Fall 2025

Daffodil International University

Assistant Professor

Dhaka, Bangladesh

Spring 2022-Summer 2022

PROJECTS

- **FlySpotter (2025-Present)**

- Developed a pipeline to analyze which parameter e.g. padding, zooming, model inference threshold increasing affect the detection accuracy from the video frame
- Created a dataset, annotate the dataset and trained our models with this dataset.
- Run real-time inference on edge devices of 4K video collected by DJI Matrice 600. Analyzing the effect of inference time, energy consumption, accuracy, precision, IoU after zooming and padding in edge devices.

- **WASP (2025-Present)**

- Proposed a model-agnostic robustness training framework for lightweight edge vision models operating under severe weather and illumination-induced distribution shifts.
- Evaluated on AgriPest-13 and IP102 across 70 physics-based corruptions using MobileNetV3 and ResNet backbones.
- Validated real-world deployment on Raspberry Pi-based edge nodes with Long Range (LoRa) communication, demonstrating robust on-device inference and energy-efficient invasive species detection.

- **Righteous (2024)**

- Developed Righteous, an edge deployment right-sizing tool for simulators and testbeds to tunable hardware, software, and network parameters.
- Proposed Informed Pareto Simulated Annealing (iPSA) to find near-optimal resource configurations based on user-defined performance goals.
- Achieved up to $3.5\times$ better configurations and $76\times$ faster optimization compared to SOTA techniques.

- **Third Eye (2023)**

- Deployed 8-bit quantized MobileNetV3-Large model to run inference faster with minimal accuracy loss on a Raspberry Pi 4B
- Built a transmitter node (Raspberry Pi 4B, Camera and LoRa) sending object labels to a receiver node. [\[Details\]](#)

- **Spotted Lantern Fly Detection through Drones (2023)**

- Trained YOLOv8 model with datasets from Penn State and the New York State Department of Agriculture and Management
- Submitted at Cornell Institute for Digital Agriculture Hackathon [\[Details\]](#)

- **Robot Task Planning with LLMs (2023)**

- Explored capabilities of LLMs like GPT-3 to interpret grounded instructions and decompose them into robot-executable subtasks. [\[Details\]](#)

SOFTWARE ARTIFACTS

- Trained a custom MobileNetV3 model to detect 4 different stages of Spotted Lantern Fly.
- Deployed in collaboration with the New York State Department of Agriculture and Markets (NYSDAM).

SCHOLARSHIPS AND AWARDS

- ACM/IEEE Symposium on Edge Computing 2024(SEC 2024) Travel Grant November 2024
- Jahangirnagar University Scholarship (5th Position in class) December 2013
- Jahangirnagar University Scholarship (7th Position in class) August 2013